# Logic, Language, and Philosophy

## Identity

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# Overview

- Identity
  - Why?
  - How?
- Translations
- Models, validity, counterexamples
- More on descriptions
  - Quantified phrases and compositionality
  - Analysing descriptions
  - Descriptions as terms
  - Frege's analysis
- Homework

# Recap

- Truth definition; interpretation quantifiers with assignments
- Validity: undecidable property
- Proving validity: reasoning about the truth definition
- Proving invalidity: constructing counterexample

# Recap validity

**Definition (Validity of argument scheme)**

An argument scheme $\phi_1 \ldots \phi_n \,/\, \psi$ is valid iff for all $M$ and $g$:
if $V_{M,g}(\phi_1) = \ldots = V_{M,g}(\phi_n) = 1$, then $V_{M,g}(\psi) = 1$

Notation: $\phi_1 \ldots \phi_n \models \psi$

# Examples

Example 1: $\forall x(\exists y Py \rightarrow Qx) \models \exists y Py \rightarrow \forall x Qx$

Proof:

- Take arbitrary $M$ and $g$ such that $V_{M,g}(\forall x(\exists y Py \rightarrow Qx) = V_{M,g}(\exists y Py) = 1$
- Then: for all $d \in D_M$ it holds that $V_{M,g[x/d]}(\exists y Py \rightarrow Qx) = V_{M,g[x/d]}(\exists y Py) = 1$ (truth clause $\forall$)
- And thus: there is a $d' \in D_M$ such that $V_{M,g[y/d',x/d]}(Py \rightarrow Qx) = V_{M,g[y/d'x/d]}(Py) = 1$ (truth clause $\exists$)
- From that it follows that: $V_{M,g[y/d',x/d]}(Qx) = 1$ (Modus Ponens)
- This holds for all $d \in D_M$, so we have: $V_{M,g[y/d']}(\forall x Qx) = 1$ (truth clause $\forall$)
- And since $y$ does not occur free in $\forall x Qx$, we have $V_{M,g}(\forall x Qx) = 1$

Example 1, with free variables: $Ax \wedge Bx \models Ax$

Proof (only the relevant steps):

- Take arbitrary $M$ and $g$ such that $V_{M,g}(Ax \wedge Bx) = 1$
- Then: $V_{M,g}(Ax) = V_{M,g}(Bx) = 1$ (truth clause $\wedge$)
- So, $V_{M,g}(Bx) = 1$

# Recap invalidity

**Definition (Invalidity of argument scheme)**

An argument scheme $\phi_1 \ldots \phi_n \,/\, \psi$ is invalid iff there is at least one $M$ and $g$ such that:
$V_{M,g}(\phi_1) = \ldots = V_{M,g}(\phi_n) = 1$ and $V_{M,g}(\psi) = 0$

Notation: $\phi_1 \ldots \phi_n \not\models \psi$

# Examples

Example 1: $\forall x(Ax \vee Bx), \exists x \neg Ax, \exists x \neg Bx, \forall x((Ax \wedge Bx) \rightarrow Cx) \not\models \exists x Cx$

Countermodel $M$:

- $D_M = \{1, 2\}$
- $I_M(A) = \{1\}, I_M(B) = \{2\}, I_M(C) = \emptyset$
- Note: no free occurrences of variables, so $g$ is irrelevant

Example 2, with free variables: $Ax \wedge Bx, \exists y \neg Ay, \exists y \neg By, (Ax \wedge Bx) \rightarrow Cx \not\models \exists y Cy$

- Since $x$ is free in $Ax \wedge Bx$ and in $(Ax \wedge Bx) \rightarrow Cx$, we also need to specify $g$ as part of the counter-example
- So: previous model + $g$: $g(x) = 1$

# Examples (cont.)

Example 3: $\forall x Rxx$ / $\forall x \forall y Rxy$

Countermodel $M$:

- $D_M = \{1, 4\}$
- $I_M R = \{\langle 1, 1\rangle, \langle 2, 2\rangle\}$

Graphically:



# Identity

# Identity

## Why?

# Other

(1) John loves someone and Bill loves someone

(2) John loves someone and Bill loves someone else

(3) $\exists x Ljx \wedge \exists y Lby$

Note: (3) is ok for (1), but not for (2).

Let $D_M = \{\mathtt{j}, \mathtt{b}, \mathtt{m}\}$, $I_M(j) = \mathtt{j}$, $I_M(b) = \mathtt{b}$, $I_M(m) = \mathtt{m}$ and $I_M(L) = \{\langle \mathtt{j}, \mathtt{m} \rangle, \langle \mathtt{b}, \mathtt{m} \rangle\}$.

Then $V_{M,g}(\exists x Ljx \wedge \exists y Lby) = 1$, since $V_{M,g[m/x]}(Ljx) = V_{M,g[m/y]}(Lby) = 1$.

# Only

(4) John loves Mary

(5) John loves only Mary

(6) Only John loves Mary

(7) *Ljm*

Note: (7) is ok for (4), but not for (5), since (7) is consistent with John loving also Sue, which is excluded by (5). And (7) is also not ok for (6), since (7) does not exclude that Mary is also loved by Bill, which is ruled out by (6).

# Implicit exhaustification

Answers to questions are naturally interpreted *exhaustively*:

(8) *A* to *B*: Do you know which students got a passing grade?

(9) *B* to *A*: Yes, I do: Wang Chong, Priscilla Desert, and Hilary de Jongh got a passing grade

*B*'s answer is naturally interpreted as:

(9') Wang Chong, Priscilla Queen, and Hilary de Jongh are the students / the only students / all the students that got a passing grade

# Numerals

(10) Mary loves someone

(11) Mary loves two people

(12) $\exists x\ Lmx$

(13) $\exists x\ Lmx \wedge \exists y\ Lmy$

Note: (13) is also true if Mary loves only one person:

Let $D_M = \{\mathtt{m}, \mathtt{j}\}$, $I_M(m) = \mathtt{m}$, $I_M(j) = \mathtt{j}$, and $I_M(L) = \{\langle \mathtt{m}, \mathtt{j} \rangle\}$. Then:

$V_{M,g}(\exists x\ Lmx \wedge \exists y\ Lmy) = 1$, since $V_{M,g[j/x]}(Lmx) = V_{M,g[j/y]}(Lmy) = 1$

# Descriptions

(14) The King of France is bald

(15) $\exists x(KFx \wedge Bx)$

Note: (15) is not ok for (14) since it does not exclude that there is more than one King of France. And worse:

Let $D_M = \{\mathtt{b}, \mathtt{j}\}$, $I_M(b) = \mathtt{b}$, $I_M(j) = \mathtt{j}$, $I_M(KF) = \{\mathtt{b}, \mathtt{j}\}$, $I_M(B) = \mathtt{j}$. Then:

$V_{M,g}(\exists x(KFx \wedge Bx)) = 1$, since $V_{M,g[j/x]}(KFx \wedge Bx) = 1$

But also:

$V_{M,g}(\exists x(KFx \wedge \neg Bx)) = 1$, since $V_{M,g[b/x]}(KFx \wedge \neg Bx) = 1$

# Identity

## How?

# Adding identity

Syntax:

- Add a 2-place constant '$=$' to the language

Examples:

- $a = b$, $x = y$, $a = x$
- Note: instead of $\neg(s = t)$, we write: $s \neq t$

Semantics:

- In all models $M$:

$$I_M(=) = \{\langle d, d\rangle \mid d \in D\}$$

In words: identity is the relation that every object has to itself and only to itself.

# Identity in a model



# Expressive power

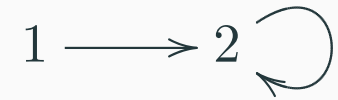
Adding $=$ extends expressive power:

Compare:

$M_1$:




$M_2$:




Note:

(a) $\exists x \exists y Rxy$: true in both $M_1$ and $M_2$

(b) $\exists x \exists y (x \neq y \wedge Rxy)$: true in $M_2$ but false in $M_1$

In other words: $\{M | M \models b\} \subset \{M | M \models a\}$, i.e., $b \models a$ and $a \not\models b$

# Translations

# Other

(16) John loves someone and Bill loves someone
$\exists x Ljx \wedge \exists y Lby$

(17) John loves someone and Bill loves someone else
$\exists x \exists y (Ljx \wedge Lby \wedge x \neq y)$

# Only

(18) John loves Mary
$Ljm$

(19) John loves only Mary
$Ljm \wedge \forall x(Ljx \rightarrow x = m)$

(20) Only John loves Mary
$Ljm \wedge \forall x(Lxm \rightarrow x = j)$

# Combining only's

"Only John loves only Mary":

- $Ljm \wedge \forall x(Ljx \rightarrow x = m) \wedge \forall x(Lxm \rightarrow x = j)$

which is equivalent to:

- $Ljm \wedge \forall x \forall y(Lxy \rightarrow (x = j \rightarrow y = m) \wedge (y = m \rightarrow x = j))$

which is equivalent to:

- $Ljm \wedge \forall x \forall y(Lxy \rightarrow (x = j \leftrightarrow y = m))$

# Numerals

(21) There is at least one $P$
$\exists x Px$

(22) There is at most one $P$
$\forall x \forall y((Px \wedge Py) \rightarrow x = y)$

(23) There is exactly one $P$
$\exists x(Px \wedge \forall y(Py \rightarrow x = y))$

# Numerals (cont.)

(24) There are at least two $Q$s

$$\exists x \exists y (x \neq y \wedge Qx \wedge Qy)$$

(25) There are at most two $Q$s

$$\forall x \forall y \forall z ((Qx \wedge Qy \wedge Qz) \rightarrow (x = y \vee x = z \vee y = z))$$

(26) There are exactly two $Q$s

$$\exists x \exists y (x \neq y \wedge Qx \wedge Qy \wedge \forall z (Qz \rightarrow (z = x \vee z = y)))$$

# Descriptions

(27) The King of France is bald
$\exists x(KFx \wedge \forall y(KFy \rightarrow x = y) \wedge Bx)$

Three elements:

1. Existence: $\exists xKFx$
2. Uniqueness: $\forall y(KFy \rightarrow x = y)$
3. Baldness: $Bx$

# Compressed formulations

Sometimes translations can be stated in a more compressed form, e.g.:

(28) John loves only Mary:
$Ljm \wedge \forall x(Ljx \rightarrow x = m) \Leftrightarrow \forall x(Ljx \leftrightarrow x = m)$

(29) King of France is bald:
$\exists x(KFx \wedge \forall y(KFy \rightarrow x = y) \wedge Bx) \Leftrightarrow \exists x \forall y((KFy \leftrightarrow x = y) \wedge Bx)$

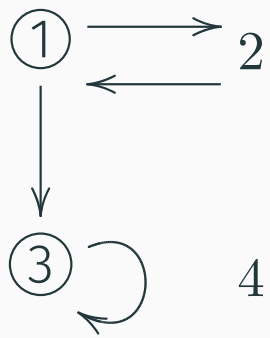
(30) There is exactly one P:
$\exists x(Px \wedge \forall y(Py \rightarrow x = y)) \Leftrightarrow \exists x \forall y(Py \leftrightarrow x = y)$

# Models, validity, counterexamples

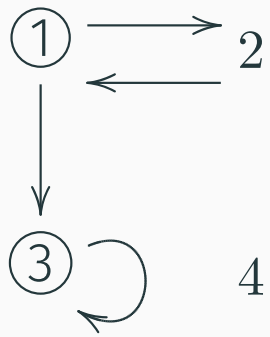
# Model checking

Let $M$ be
$D_M = \{1, 2, 3, 4\}, I_M(A) = \{1, 3\}, I_M(R) = \{\langle 1, 2\rangle, \langle 2, 1\rangle, \langle 1, 3\rangle, \langle 3, 3\rangle\}$.

$M$ can be represented by the following graph:



# Examples




a. $\forall x(Ax \rightarrow \exists y Ryx)$

b. $\forall x(Ax \rightarrow \exists y(x \neq y \wedge Ay \wedge Ryx))$

c. $\exists x(\neg\exists y Rxy \wedge \forall z(\neg\exists y Rzy \rightarrow z = x))$

# Validity

Validity of formulas and argument schemes is done by reasoning about the truth definition.

$$\models \forall x \forall y (x = y \to (Px \to Py))$$

**Proof**:
Take arbitrary $d, d' \in D$. If $V_{M,g[x/d,y/d']}(x = y) = 1$ then $g[x/d, y/d'](x) = g[x/d, y/d'](y)$, viz., $d = d'$. Suppose $V_{M,g[x/d,y/d]}(Px) = 1$, then $V_{M,g[x/d,y/d]}(Py) = 1$ follows immediately. Since $d, d'$ were arbitrarily chosen the result follows. ■

# *Leibniz's laws of identity*

"Indiscernibility of identicals":

$$\models \forall x \forall y (x = y \to \forall X (Xx \to Xy))$$

"Identity of indiscernibles":

$$\models \forall x \forall y (\forall X (Xx \to Xy) \to x = y)$$

Both use second-order quantification.

# Counterexamples

As usual: model in which premisses are true and conclusion is false.

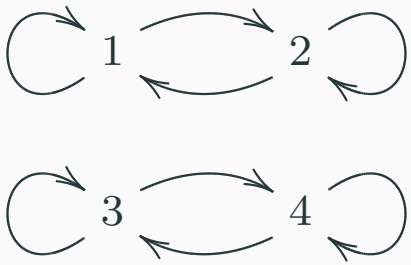
Example 1:

$$\forall x \exists y (x \neq y) \not\models \exists x \exists y \exists z (x \neq y \wedge x \neq z \wedge y \neq z)$$

That this is not a valid scheme is shown by a model $M$ with $D = \{1, 2\}$

Example 2:

$$\forall x \exists y (Rxy \wedge x \neq y), \forall x \forall y \forall z ((Rxy \wedge Ryz) \rightarrow Rxz) \,/\, \forall x \forall y (x = y \vee Rxy \vee Ryx)$$



# More on descriptions

# To be or not to be

Russell's analysis:

- descriptions *assert* existence and unicity

Frege's analysis (informal):

- descriptions *presuppose* existence and unicity

When existence and/or unicity fails:

- Russell: statement is false
- Frege: statement lacks a truth value

Key element: different views on how descriptions refer

# More on descriptions

## Quantified phrases and compositionality

# Compositionality

Compositionality (rough version):

> *The meaning (reference) of a complex expression is determined by the meanings (references) of its component parts and the way in which these are combined*

Accounts for:

- Finite characterisation of syntax and semantics of (potentially) infinite sets of expressions of a language
- Ability of language users to use and to understand new expressions

# Quantified phrases as constituents

Quantified phrases are constituents:

(31) Every man is happy

(32) Some men who like to dance are happy

(33) The man wearing a hat is happy

# Quantified phrases and compositionality

According to compositionality the meaning (reference) should be a function of the meaning (reference) of the respective constituent parts.

However:

(34) John is happy = $Hj$

(35) Every man is happy $\neq H$(every man)

Reason: whereas *John* refers to an individual *Every man* does not.

Solution in predicate logic $\approx$ Russell's analysis in OD:
treat quantifiers syncategorematically.

[Aside: quantified phrase *can* be treated as denoting expressions in higher-order logic.]

# More on descriptions

## Analysing descriptions

# Definite descriptions

What about definite descriptions?

Note:

(a) descriptions refer to individuals

(b) descriptions are complex

(a) is similarity with constants and variables, (b) is similarity with quantified phrases.

Russell: treat them in the same way as other quantified phrases.

Alternative: treat them in the same way as other individual-denoting phrases, such as constants and variables, i.e., as a third kind of *term*.

# More on descriptions

## Descriptions as terms

# The $\iota$-operator: syntax

Observation: propositional functions correspond to properties.

Basic idea: treat the definite article as an expression that turns a propositional function into an individual denoting expression.

This is done by means of a new logical expression, called the *iota-operator*:

- If $\phi$ is a formula, then $\iota x \phi$ is a term

# Examples

- $\iota x P x$ = the $P$
- $\iota x(Px \wedge Rcx)$ = the $P$ that stands in the relation $R$ to $c$

When treated as terms, descriptions operate in the same way as constants and variables:

- $Q(\iota x P x)$ = the $P$ is $Q$
- $S(a, \iota x(Px \wedge Rxc))$ = $a$ stands in the relation $S$ to the $P$ that stands in the relation $R$ to $c$
- $R(\iota x P x, \iota y Q y)$ = the $P$ and the $Q$ stand in the relation $R$

# Syntax (rev.)

Since terms are formed from formulas *and* occur in formulas, the syntax is now *a simultaneous inductive definition* of two classes of expressions, viz., that of terms and that of formulas.

# The $\iota$-operator: semantics

Interpretation:

$[\![\iota x \phi]\!]_{M,g}$ is the unique individual $d \in D_M$ such that $V_{M,g[x/d]}(\phi) = 1$

Note:

- If there is no individual that satisfies $\phi$, or if there is more than one, then $[\![\iota x \phi]\!]_{M,g}$ is undefined
- If the interpretation of a subexpression of an expression is undefined, then so is the interpretation of the expression itself (compositionality)

*That means that bi-valence, i.e., $\phi \vee \neg\phi$, fails as a general principle*

# More on descriptions

## Frege's analysis

# The nil-element

Basic features:

- Lack of reference is a shortcoming of natural language [recall *Begriffsschrift]*
- A proper (logical) language always supplies a referent for any name (including descriptions)
- For descriptions that do not refer, this is a distinguished 'nil-element' $d_0$

Saving bi-valence:

- Define $[\![\iota x \phi]\!]_M$ as the unique individual $d \in D_M$ such that $V_{M,g[x/d]}(\phi) = 1$ if such $d$ exists; and $d_0$ otherwise
- Exclude $d_0$ from the interpretation of individual constants and $n$-ary predicate constants.

# Observations

- This is a purely formal solution, but in a formal context not uncommon
- Formality is an objection only in the context of natural language
- In the context of natural language giving up bi-valence should be an option

# Homework

# Homework

Homework

- Read chapter 16 and make exercises 1(a), 2(a,d), 4(a,b(ii,iii)), 5(c,d), 6(a,d)